



Glacial lakes in the Koshi, Gandaki, and Karnali river basins of Nepal, the Tibet Autonomous Region of China, and India

The data on glacial lakes in 2015 were derived through a semi-automatic method using Landsat imagery from 2015 to 2016. The Normalized Difference Water Index threshold value was used for classification. Further, the lake boundary was revised or manually updated by incorporating those lakes that were missed because of atmospheric and physical processes such as being frozen, covered under snow, or obstructed by shadows and clouds. The mapped glacial lakes were validated and modified where necessary by overlaying the Landsat imagery over previous inventory datasets. They were also cross-checked with high-resolution Google Earth imagery. The attributes of glacial lakes were derived from SRTM DEM, and the data were projected in the WGS UTM projection system. This dataset was prepared to support the Green Climate Fund project proposal formulated by UNDP Nepal.

Simple

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Purpose	The dataset provides the status of glacial lakes in 2015 in three main river basins (Koshi, Gandaki, and Karnali, including Mahakali) that drain into Nepal. It is useful for researchers, scientists, and infrastructure developers, and policymakers for glacial water resource management, climate change impact studies, and glacial hazard risk reduction and mitigation.
Status	Completed
Maintenance and update frequency	As needed
(Theme)	• Glacial Lake • Potential dangerous glacial lakes • GLOF • Cryosphere
Keywords (Place)	• Nepal • China • India
Use limitation	Free to use with attribution to the source. Suggested citation: ICIMOD. (2020). Glacial lakes in the Koshi, Gandaki, and Karnali river basins of Nepal, the Tibet Autonomous Region of China, and India [Data set]. ICIMOD. https://doi.org/10.26066/RDS.1971946
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Spatial representation type	Vector
Metadata language	eng
Character set	UTF8
Topic category	• Geoscientific information

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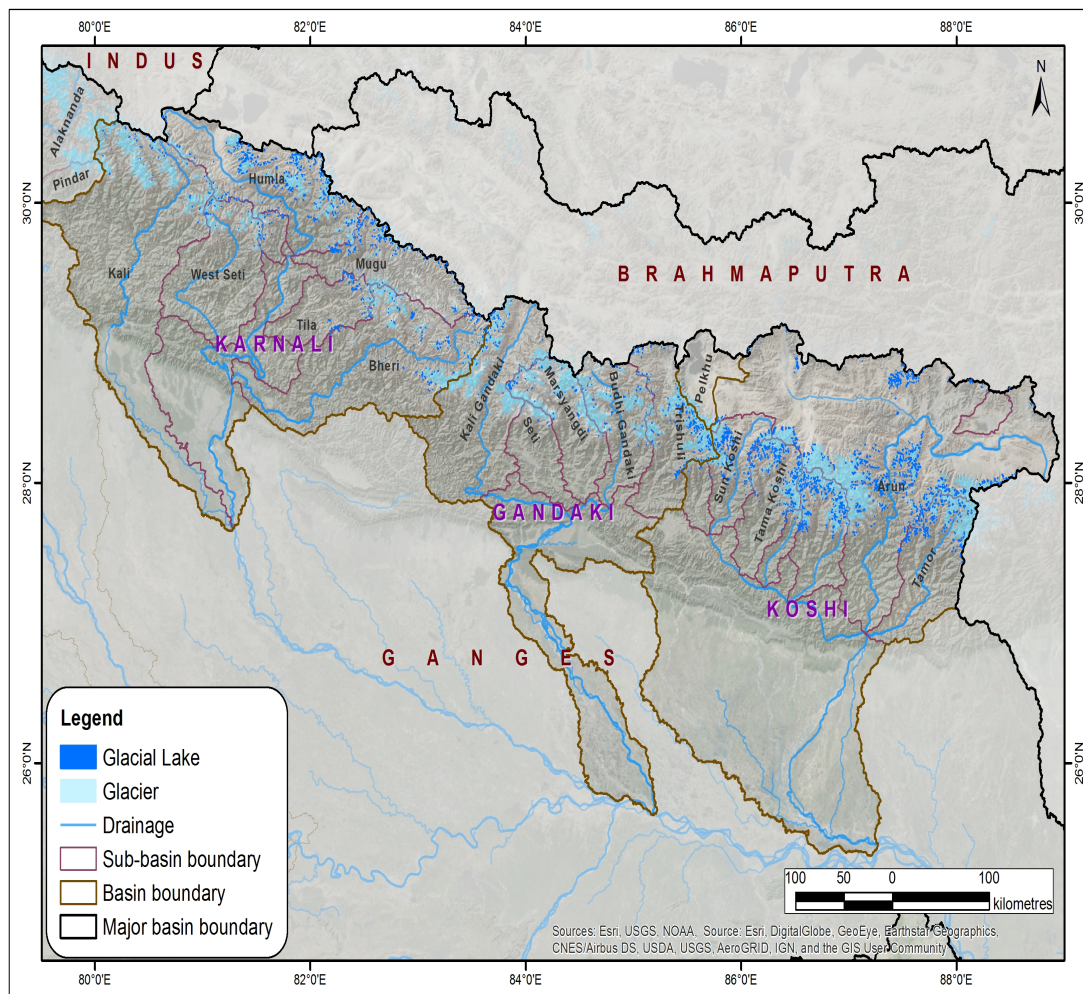
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Reference system identifier	WGS 1984
Distributor <i>ICIMOD -</i>	
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Hierarchy level	Dataset
File identifier	777e6cb6-c2db-47ab-852c-48bd0f24a158 XML
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